

CHONGKAI (LUCAS) LYU

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RESEARCH INTERESTS

Communication and sensing systems; digital signal processing and modulation; wireless and mobile sensor networks; embedded sensing systems; detection and estimation; and machine learning for communications.

EDUCATION

University of British Columbia

B.A.Sc. in Computer Engineering (Co-op); Minor in Commerce

Vancouver, BC

Expected May 2027

Cumulative Average: 83%

Selected coursework with grades:

- Linear Algebra for Engineers (97%)
- Machine Learning with Engineering Applications (91%)
- Circuit Analysis (100%)
- Operating Systems (93%)
- Engineering Analysis II (92%)
- Digital Systems Design (94%)
- Computer Engineering Design Studio II (95%)
- Computer Graphics (90%)

SELECTED RESEARCH AND ENGINEERING PROJECTS

DDS and Nios II Modulation System (FPGA) [🔗 Git Repo](#) | UBC CPEN

June 2024

Verilog/SystemVerilog, DE1-SoC, DDS LUTs, ASK/FSK/BPSK/QPSK, LFSR, Nios II, Qsys/Avalon, clock-domain crossing, I2S/VGA, testbenches

- Designed and implemented an FPGA **direct digital synthesis (DDS)** and digital-modulation system on the **DE1-SoC** in **Verilog/SystemVerilog**, generating sine, cosine, and sawtooth waveforms from lookup tables with configurable phase increments
- Implemented **ASK**, **FSK**, **BPSK**, and **QPSK** digital-modulation paths—including in-phase/quadrature composition for QPSK—driven by a 5-bit **LFSR** pseudo-random data source and a configurable clock divider
- Integrated a **Nios II** soft-core through **Qsys/Avalon** so C firmware controls the modulation and peripheral logic, and designed parameterized fast-to-slow and slow-to-fast **clock-domain** transfer modules to synchronize data across clock domains
- Verified waveform generation and modulation with **SystemVerilog** testbenches, and exercised the design on hardware with **I2S** audio output and VGA oscilloscope signal visualization

AquaWorks: Water-Quality Monitoring Robot

June 2026 – January 2027

Project Co-Lead | University of British Columbia

STM32, GPIO/PWM, ESC/motor control, Bluetooth, water-quality sensors, C/firmware, system architecture

- Co-founded and currently co-lead an interdisciplinary UBC team developing a modular **surface water-quality monitoring robot**, coordinating mechanical, hardware, software, and business sub-teams from system definition through integration and testing
- Architected an **STM32**-controlled floating platform integrating bidirectional propulsion (ESC/motors via GPIO/PWM), modular water-quality sensors (turbidity, conductivity, temperature, pH, dissolved oxygen), wireless (**Bluetooth**) communication, and location/time-linked measurement logging
- Developing and debugging **STM32** firmware and embedded integration for sensor interfaces, power distribution, propulsion control, wireless communication, and waterproof electronics
- Defined phased bench and controlled-water validation for buoyancy, propulsion, sensor calibration, and planned communication-reliability testing; co-prepared an approximately **CAD 4,850** AMS Sustainability Projects Fund proposal

Multi-threaded Audio Processing & DSP UBC CPEN

September 2023 – December 2023

Java, concurrency, Discrete Fourier Transform (DFT), echo filtering, JavaFX

- Designed a multi-threaded client-server audio-processing application in **Java** with thread-pool request handling and synchronized access to shared processing state
- Implemented a **DSP** pipeline with **DFT**-based frequency-domain analysis, echo filtering, and real-time waveform visualization in a JavaFX interface

Hand Gesture Recognition [Git Repo](#)

September 2024 – December 2024

Python, MediaPipe, OpenCV, scikit-learn, PyTorch, TensorFlow, experiment design

- Built an end-to-end gesture-classification pipeline over a self-collected **6,000+**-sample dataset, extracting 21 **MediaPipe** hand landmarks as normalized geometric features with reproducible stratified train/test splits
- Evaluated classical classifiers (**KNN**, random forest, **SVM**, logistic regression, MLP) and **PyTorch/TensorFlow** CNNs across landmark- and image-based pipelines using accuracy, precision/recall/F1, confusion matrices, and 10-fold cross-validation, and deployed real-time **OpenCV** camera inference

iPod Music Player with PicoBlaze Processor (FPGA) [Git Repo](#) | UBC CPEN

June 2024

Verilog, DE1-SoC, PicoBlaze soft-core, I2S audio, SDRAM/Flash, LCD, PS/2

- Built an **FPGA** music player on the **DE1-SoC** in **Verilog** with **I2S** audio output, Flash/SDRAM audio streaming, PS/2 keyboard control, and bidirectional/variable-speed playback, then extended it with a **PicoBlaze** soft-core (ALU, instruction decode, register file, stack)
- Used the soft-core to drive an LCD oscilloscope display for real-time audio-waveform rendering, integrating processor, memory, and peripheral interfaces in a hardware-software design

RELEVANT PROFESSIONAL EXPERIENCE

Software Engineer

January 2026 – August 2026

Korotu Technology: FarmWatch Agroforestry Platform | [Website](#)

Vancouver, BC

- Developed environmental decision-support software in **Python/Django** with layered handler/service/repository design, computing carbon-stock, biodiversity-potential, and soil-stability KPIs by overlaying multi-source geospatial layers (Sentinel-2 NDVI, land cover, water proximity) on field geometries with **PostGIS**, **GeoPandas**, and **rasterio**
- Reduced repeated field-analysis time by approximately **60%** through persisted metric caching and per-field recomputation locks, and parallelized multi-field computation with up to **8** workers to build reproducible data-processing pipelines

TECHNICAL SKILLS

Signal Processing & Communications: DDS, DFT and frequency-domain analysis, digital modulation (ASK/FSK/BPSK/QPSK), waveform generation, LFSR sequences, echo filtering, I2S, signal visualization

Embedded & Digital Systems: FPGA/DE1-SoC, Verilog, SystemVerilog, Nios II, PicoBlaze, STM32, ESP32, Arduino, FSM design, clock-domain synchronization, UART/I2C/SPI, Bluetooth

Programming & Scientific Computing: C, C++, Python, Java, multi-threading and concurrency

Machine Learning: PyTorch, TensorFlow, OpenCV, MediaPipe, KNN, random forest, SVM, neural networks and CNNs

Tools & Platforms: Qsys/Platform Designer, ModelSim/QuartaSim, SystemVerilog testbenches, Git, Linux, Docker